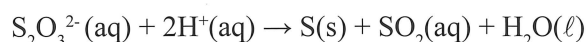


Experiment 39: Reactant concentration and rate of reaction

A study of the factors that influence rates of reaction is important in developing an understanding of how environmental, biological and industrial processes proceed. The luminol test reactions for example occur quite quickly and with dramatic effect. Different chemical reactions occur at different rates. Some reactions, such as many neutralisation and precipitation reactions, occur very rapidly. Other reactions, such as the air oxidation of vitamin C and the rusting of iron, are relatively slow. In this experiment you will investigate how one factor, concentration, can affect the rate of reaction.

The reaction between sodium thiosulfate and hydrochloric acid can be represented by the equation:



The time taken for the precipitate of sulfur to obscure a cross marked on a piece of paper placed under the reaction flask is used as a measure of the reaction rate. The less time taken for the cross to 'disappear', the faster is the rate of the reaction. In this activity you will investigate the rate of this reaction as a function of the sodium thiosulfate concentration.

Equipment

sodium thiosulfate solution [$\text{Na}_2\text{S}_2\text{O}_3$] 0.25 mol L⁻¹ (150 mL)
hydrochloric acid [HCl] 2 mol L⁻¹ (30 mL)
distilled water
graduated cylinders (10 mL, 25 mL and 100 mL)
conical flasks (two 100 mL)
stop clock

SAFETY NOTE:

- Avoid breathing in any sulfur dioxide that is evolved while you are waiting for the dot to disappear. Do not carry out this experiment on the bench if you are asthmatic.

Procedure

- Copy the table below into your laboratory notebook and record your results directly into it.

Vol. of 0.25 mol L ⁻¹ Na ₂ S ₂ O ₃ (mL)	Vol. of H ₂ O added (mL)	Total volume after adding & mixing 5 mL HCl	Concentration of Na ₂ S ₂ O ₃ on mixing (mol L ⁻¹)	Time for cross to disappear (s)	$\frac{1}{\text{Time (s}^{-1}\text{)}}$
45	0	50	0.225		
35	10	50			
25	20	50			
15	30	50			
5	40	50			

Notes